

Sarweshwaran R S

Software Engineer | Integrated Dual Degree (B.Tech + M.Tech) | Software Engineering (VIT)
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Professional Summary

Software Engineering postgraduate with strong foundations in Data Structures, Object Oriented Programming (OOP), Design Principles. Built 4+ full-stack and ML-based applications using Java, Spring Boot, and React, focusing on scalable backend systems and real-time processing. Passionate about solving real-world problems using efficient, high-performance systems across the full software development lifecycle.

EDUCATION

Vellore Institute of Technology (VIT), Vellore.
Integrated M.Tech – Software Engineering (2021-2026)
CGPA: 8.34 / 10

Lakshmi Garden Matric Higher Secondary School, Vellore.
Class XII (2020-2021) – 89.8%.

Seventh Day Adventist Matriculation Higher Secondary School, Vellore.
Class X (2018-2019) – 92.0%.

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, JavaScript

Frontend: HTML5, CSS3, React.js, Next.js

Backend: Node.js, Express.js, Spring Boot, REST API's

Core CS: Data Structures & Algorithms, Object Oriented Programming, Object Oriented Design, Design Patterns.

Tools: Docker, Git, GitHub, Postman, Insomnia

ML/AI: PyTorch, TensorFlow / Keras, Scikit-learn

Databases: MySQL, MSSQL, PostgreSQL, MongoDB.

OS: Windows, Ubuntu, Linux

PROJECTS

Real-Time Video Conferencing Application

- Developed a scalable, secure video conferencing platform enabling the real-time communication and user interaction.
- Implemented authentication, session handling, and low-latency streaming using third-party APIs.
- Designed responsive UI and ensured seamless cross-device usability.

AI-Based Plant Disease Detection System (Generative AI)

- Designed a deep learning model for plant disease classification using image data, implemented preprocessing, model training, and optimization for improved accuracy.

- Tech: Python, PyTorch, Scikit-learn, Kaggle

Full Stack Project Management System (Java + Spring Boot)

- Developed a scalable task management system supporting multi-user workflows.
- Designed REST APIs and implemented CRUD operations with relational database integration.
- Built responsive frontend for seamless user interaction.
- Tech: Java, Spring Boot, React.js, MySQL.

Microservices-Based Backend System (Spring Boot Architecture)

- Designed distributed backend using microservices architecture with API Gateway.
- Implemented inter-service communication and module service design.
- Containerized services for scalable deployment.
- Tech: Java, Spring Boot, Docker, Kafka, gRPC.

PUBLICATIONS

Journal Submission

- “TriSpectraNet: Bidirectional Cross-Attention Fusion With Spectral Encoding and Temporal Smoothing for Industrial Acoustic Anomaly Detection”.
- Submitted to International Journal of Computational Intelligence Systems (**Springer Nature**) – April 2026.

Book Chapter

Plant Disease Detection Using Generative AI and Deep Learning Models

- Published in Humans and Generative AI Tools for Collaborative Intelligence, IGI Global, 2025.
- DOI: 10.4018/979-8-3693-8332-2. ch009 [[Link](#)]

CERTIFICATIONS

Introduction to Generative AI – Google [[Link](#)]

Microsoft Azure AI – 900 [[Link](#)]

Ethical Hacking (Web Applications) – Udemy [[Link](#)]